ModelIt!

Student Activity Pack

Why Do Pandemics Spiral Out of Control?

Modeling Infectious Disease Transmission and Public Health Interventions

Grade 11L1 | NGSS HS-LS2-2, HS-LS4-6 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved.

Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

ModelIt!

What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What do you think determines whether a disease outbreak stays local or becomes a global pandemic?**

2. **What does 'exponential growth' mean, and can you give an example of how it applies to disease spread?**

3. **Why do you think public health officials sometimes recommend extreme measures (lockdowns, travel bans) when there are only a few cases?**

4. **What is herd immunity, and why do different diseases require different levels of vaccination coverage?**

2 Scoring Rubric

ModelIt!

Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

ModelIt!

What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

R0 Value | Vaccination Rate | Susceptible Population | Infection Rate | Recovery Rate | Effective R (Re)

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

ModelIt!

Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

ModelIt!

Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Uncontrolled Outbreak

R0: 2.5 | Vaccination: 0% | Interventions: None

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Vaccination Shield

R0: 2.5 | Vaccination: 60% | Interventions: None

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Timing Is Everything

R0: 2.5 | Social distancing at Day 10 vs. Day 30

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

ModelIt!

Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Superspreader Events — Occasions where a single infected person transmits to an unusually large number of others — research shows 10-20% of infected people cause 80% of transmission, making disease spread far more uneven than R0 alone suggests

Hospital Capacity — The number of ICU beds, ventilators, and healthcare workers available — when infection rate exceeds hospital capacity, mortality spikes because treatable patients cannot receive care, creating a catastrophic amplification of the death toll

Behavioral Compliance — The percentage of the population that actually follows recommended interventions like social distancing and mask-wearing — compliance typically starts high during peak fear and declines over time, creating waves of re-emergence as Re fluctuates above and below 1

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

ModelIt!

The Big Question

Why do some outbreaks burn out on their own while others explode into global pandemics — and what determines whether we can stop them? Use evidence from your simulation to support your answers.

1. **Why does exponential growth make early-stage pandemics so deceptively dangerous — and why do people consistently underestimate the threat until it's too late?**

2. **How does the herd immunity threshold equation (1 - 1/R0) explain why measles outbreaks occur even in highly vaccinated populations?**

3. **Why do epidemiologists say 'if it looks like you overreacted, you probably did it right' — what does this reveal about the paradox of successful early intervention?**

4. **How could your model be extended to account for superspreader events, which make disease spread far more uneven than a simple R0 value suggests?**

5. **What are the ethical trade-offs between implementing early, aggressive interventions (which cause economic harm but save lives) and waiting for more data (which costs lives but reduces economic disruption)?**

8 STEM Challenge: Design a Pandemic Response Plan

ModelIt!

YOUR ENGINEERING MISSION

Design an evidence-based pandemic response plan for a novel respiratory pathogen, specifying intervention triggers, timing, and resource allocation to minimize both infections and economic disruption.

THE SCENARIO

A novel respiratory virus with R0=3.0 has been detected in a city of 2 million people. Initial cases are doubling every 4 days. No vaccine exists yet, but one could be available in 6 months. Your public health team must design a response plan that specifies when to implement social distancing, testing and contact tracing protocols, and resource allocation for hospitals — with the constraint that economic shutdown costs $500 million per month.

Engineering Design Thinking

1. WHAT EPIDEMIOLOGICAL DATA TRIGGERS EACH PHASE OF YOUR RESPONSE PLAN?

2. HOW WOULD YOU ALLOCATE LIMITED TESTING CAPACITY TO MAXIMIZE THE EFFECTIVENESS OF CONTACT TRACING?

3. WHAT IS THE MINIMUM LEVEL OF SOCIAL DISTANCING NEEDED TO KEEP RE BELOW 1 FOR THIS PATHOGEN?

4. HOW WOULD YOU COMMUNICATE THE NECESSITY OF EARLY INTERVENTION TO A SKEPTICAL PUBLIC THAT SEES ONLY A FEW CASES?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Epidemiologists study disease patterns and design public health interventions, working at the CDC, WHO, state health departments, and pharmaceutical companies, earning $75,000-$140,000/year. Mathematical Modelers who build computational disease transmission models earn $80,000-$160,000/year and are among the most in-demand scientists during outbreaks. Public Health Emergency Managers coordinate pandemic response logistics and resource allocation, earning $70,000-$130,000/year.